

ADAM

AUTONOMOUS DESKTOP AI MODULE

AI Desk Pet — Complete Component Reference & Technical Guide

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What's New in v3.0: Complete re-architecture for v32 — ILI9341 TFT display driven by a dedicated Raspberry Pi Pico, RCWL-0516 radar replacing VL53L0X, stereo INMP441 mics, PCF8574T I2C expander for touch, updated pinouts throughout. Two-zone split (Body/Head) with Wi-Fi Relay or Full Hardwired communication flows.

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1. SYSTEM OVERVIEW & ARCHITECTURE

ADAM v32 is an AI desk pet powered by the Gemini Live API, split into two independent hardware zones (Body and Head) to eliminate DMA bottlenecks and minimise the wiring through the mechanical neck joint. The Pi Zero 2W remains the AI brain in the body; a dedicated Raspberry Pi Pico in the head drives the ILI9341 TFT face display at 30 FPS independently of the AI workload.

BODY ZONE	HEAD ZONE
• Pi Zero 2W — Main AI Brain (Gemini, OpenCV, Audio) • INMP441 x2 — Stereo I2S Microphones • MAX98357A + 3W Speaker — I2S Audio Out • RCWL-0516 — Microwave Radar Presence/Wake • MG90S Pan Servo — Left/Right Neck Movement • 5V Buck Converter (LM2596/MP1584) — Power Supply	• ESP32-CAM — Video Capture + Touch I/O + Tilt Servo • Raspberry Pi Pico — Dedicated Face Renderer @ 30 FPS • ILI9341 TFT 2.42" — ADAM's Expressive Face Display • PCF8574T I2C Expander — Groups 4 Touch Sensors • TTP223 x4 — Capacitive Touch Sensors • MG90S Tilt Servo — Up/Down Head Movement

Communication Flows

FLOW 1 — Wi-Fi Relay (Recommended)	FLOW 2 — Full Hardwired (Offline)
Video: ESP32-CAM → Pi Zero via Wi-Fi UDP Emotions: Pi Zero → ESP32-CAM via Wi-Fi → Pico via GPIO4 TX wire Neck wires: Only 2 wires through joint (5V + GND) Recommended for most builds — cleaner, fewer wires	Video + Commands: ESP32-CAM ↔ Pi Zero via High-Speed UART Emotions relay: ESP32-CAM → Pico via local GPIO4 TX wire Neck wires: 4 wires (5V, GND, Pi TX, Pi RX) Offline/no-Wi-Fi variant — full hardwired

Physical Assembly Layers

Zone	Layer	Contents
HEAD	Face Panel	ILI9341 TFT + Pico + ESP32-CAM + PCF8574T + TTP223 x4
NECK	Neck Joint	MG90S Pan Servo (body-side) + MG90S Tilt Servo (head-side) + 2 or 4 wires
BODY TOP	Electronics	Pi Zero 2W + MAX98357A Amp
BODY MID	Power	2S LiPo + Buck Converter + Power PCB
BODY BOT	Audio	3W Speaker (bottom-firing) + RCWL-0516 Radar

2. BILL OF MATERIALS — QUICK REFERENCE v3.0

All prices in Indian Rupees (INR). Single-unit pricing. v3.0 adds: Raspberry Pi Pico, ILI9341 TFT display, PCF8574T I2C expander, RCWL-0516 radar, second INMP441 mic, 2S LiPo battery + buck converter. Removes: SSD1309 OLED, VL53L0X ToF sensor, USB-C adapter (replaced by LiPo battery system).

ID	Component	Category	Status	Qty	Unit (Rs.)	Total (Rs.)	Source
C01	Raspberry Pi Zero 2W (with headers)	Main Brain	ORIGINAL	1	1,099	1,099	robu.in
C02	ESP32-CAM Module (OV2640)	Vision/Head	ORIGINAL	1	349	349	robu.in
C03	Raspberry Pi Pico	Graphics MCU	NEW v32	1	349	349	robu.in
C04	2.42" ILI9341 TFT (240x320 SPI)	Display	NEW v32	1	349	349	robu.in
C06	MG90S Metal Gear Micro Servo	Neck	ORIGINAL	2	179	358	robu.in
C07	TTP223 Touch Sensor	Touch Input	ORIGINAL	4	45	180	robu.in
C07b	PCF8574T I2C Expander	Touch I/O	NEW v32	1	89	89	robu.in
C08	INMP441 I2S Microphone	Audio In	UPDATED	2	149	298	robu.in
C09	MAX98357A I2S DAC+Amp	Audio Out	SELECTED	1	219	219	robu.in
C11	3W 4-Ohm Speaker	Audio Out	ORIGINAL	1	79	79	robu.in
C12	RCWL-0516 Microwave Radar	Presence	NEW v32	1	89	89	robu.in
C13	5V Buck Converter (LM2596/MP1584 3A)	Power	UPDATED	1	149	149	robu.in
C14	2S LiPo 7.4V 2000mAh Battery	Power	NEW v32	1	599	599	amazon.in
C15	Power Mgmt (Caps+Fuse+Diode)	Power	ORIGINAL	1 set	120	120	robu.in
C16	32GB Class10/A1 Micro-SD Card	Storage	ORIGINAL	1	249	249	amazon.in
C17	PLA+ Filament (1 print, ~300g)	Enclosure	ORIGINAL	~	350	350	amazon.in
C18	Heat Sinks (Pi Zero + ESP32 + Pico)	Thermal	UPDATED	1 set	79	79	amazon.in
M01	Black Acrylic Sheet 3mm (A4)	Enclosure	ORIGINAL	1	180	180	amazon.in
M02	M3 x4mm OD Brass Inserts (x20)	Hardware	ORIGINAL	1 set	109	109	amazon.in
M03	M3 Countersunk Screws — 5 pcs	Hardware	ORIGINAL	1 set	49	49	amazon.in
M06	Misc Consumables (solder, wire, flux)	Misc	ORIGINAL	1 set	150	150	local

TOTAL (v3.0 BOM)

Rs. 5,143

NOTE: C09 (MAX98357A) is the SELECTED audio amplifier. C03 (Pico) + C04 (ILI9341 TFT) replace the SSD1309 OLED for GPU-offloaded 30 FPS face rendering. C12 RCWL-0516 radar replaces VL53L0X — simpler digital GPIO wake, no I2C polling needed. C14 (2S LiPo + C13 buck converter) replaces USB-C adapter for untethered operation.

3. POWER BUDGET & COST DISTRIBUTION

CRITICAL: Both MG90S servos MUST be powered from an ISOLATED 5V rail with 470uF capacitor banks placed physically close to each servo connector. Never draw servo current from Pi or ESP32 pins — stall spikes (up to 500mA each) WILL cause brownout resets and SD card corruption.

Power source: 2S LiPo (7.4V) → LM2596/MP1584 5V Buck Converter (3A minimum). THE GOLDEN RULE: All grounds — Pi, ESP32-CAM, Pico, Amplifier, Servos, and Buck Converter — MUST share a common GND. Without a shared ground, I2S audio and UART serial will fail.

Component	Rail	Typical (mA)	Peak (mA)	Note
Pi Zero 2W (Body)	5V Main Rail	400	1000	Wi-Fi TX peaks
ESP32-CAM (Head)	5V Head Rail	180	300	Wi-Fi TX peaks
Raspberry Pi Pico (Head)	5V → VSYS	50	100	Graphics rendering
ILI9341 TFT (Head)	Pico 3V3(OUT)	25	35	CRITICAL: 3.3V only — never 5V
INMP441 Mic x2 (Body)	Pi 3.3V ONLY	3	4	CRITICAL: 3.3V from Pi — never 5V
PCF8574T Expander (Head)	ESP32 3.3V	4	10	I2C logic at 3.3V
TTP223 x4 (Head)	ESP32 3.3V	6	6	Via PCF8574T power rail
MG90S Pan Servo (Body)	5V ISOLATED RAIL	120	500	CRITICAL: Isolated — 470uF cap
MG90S Tilt Servo (Head)	5V ISOLATED RAIL	120	500	CRITICAL: Isolated — 470uF cap
MAX98357A + Speaker (Body)	5V Main Rail	200	700	Full volume 3W
RCWL-0516 Radar (Body)	5V Main Rail	3	5	Has internal 3.3V reg
Heat Sinks (passive)	—	0	0	Passive — no power draw
TOTAL (excl. servos)	5V	891	2160	Use 3A buck converter

Neck passthrough wiring: Run 22AWG wires from body Main 5V Rail through the mechanical neck to create the Head 5V Rail. Flow 1 (Wi-Fi): 2 wires only (5V + GND). Flow 2 (Hardwired): 4 wires (5V, GND, Pi TX GPIO14, Pi RX GPIO15).

4. COMPONENT DETAILS

C01 Raspberry Pi Zero 2W (with pre-soldered headers)

ORIGINAL

Main Brain — Quad-Core ARM Cortex-A53 @ 1GHz, 512MB LPDDR2, Built-in Wi-Fi & BT

Processor	Quad-core ARM Cortex-A53 @ 1 GHz
RAM	512 MB LPDDR2 — sufficient for all ADAM v32 subsystems
Wireless	2.4 GHz 802.11 b/g/n, Bluetooth 4.2/BLE — Wi-Fi for Gemini API
GPIO	40-pin header (I2C, SPI, UART, PWM, I2S) — pre-soldered required
Power	~400 mA typical / ~1000 mA peak — from 5V Main Rail (buck converter)
Heat Sink	14x14mm self-adhesive aluminium (C18) — required for sustained Gemini loads

ROLE IN ADAM v32: Central AI brain. Runs Raspberry Pi OS Lite. Hosts: Gemini Live API voice session, OpenCV face tracking, pan servo control (GPIO12), RCWL-0516 radar wake (GPIO16), I2S stereo mic input, I2S audio output, Wi-Fi/UART link to ESP32-CAM, and persistent memory (adam_memory.json).

v32 WIRING CHANGES: I2C bus (GPIO2/3) now free — OLED and VL53L0X removed. Pan servo moved to GPIO12 (was GPIO9). Tilt servo moved to ESP32-CAM GPIO13 (off Pi entirely). UART on GPIO14(TX)/GPIO15(RX) for Flow 2 hardwired mode. RCWL-0516 on GPIO16.

Price (INR): Rs. 1,099 - Rs. 1,299 — robu.in

C02 ESP32-CAM Module (OV2640 Camera)

UPDATED

Head Vision & I/O Hub — Video Streaming + Touch Reading + Tilt Servo

Processor	ESP32 dual-core @ 240 MHz, 4MB Flash + 8MB PSRAM
Camera	OV2640 — 2MP, up to 1600x1200, JPEG hardware encoder
Wireless	2.4 GHz Wi-Fi + Bluetooth — UDP stream (Flow 1) or UART (Flow 2) to Pi
Power	5V, ~180 mA typical / ~300 mA TX peak — from Head 5V Rail
Role v32	Video capture + PCF8574T touch I/O (I2C: GPIO14/15) + Tilt servo (GPIO13) + Pico relay (GPIO4)

ROLE IN ADAM v32: Head I/O controller. Streams JPEG face frames to Pi. Reads 4 touch sensors via PCF8574T expander. Controls tilt servo on GPIO13. Relays emotion commands from Pi to Pico via software UART on GPIO4.

PAN INVERSION FIX (v25+): ESP32-CAM feed is mirrored. Pan angle = $150 - (cx_norm \times 120)$. Without this inversion ADAM tracks in the wrong direction.

Price (INR): Rs. 349 - Rs. 499 — robu.in

C03 Raspberry Pi Pico

NEW v32

Dedicated Graphics MCU — ADAM Face Renderer @ 30 FPS

Processor	RP2040 dual-core ARM Cortex-M0+ @ 133 MHz, 264KB SRAM, 2MB Flash
Interface	SPI (GP18/19/17/16/20) → ILI9341 TFT display
UART RX	GP1 — receives emotion commands from ESP32-CAM GPIO4
Power	5V → VSYS (Pin 39), outputs 3V3(OUT) Pin 36 to power TFT display
Firmware	MicroPython — renders emotion animation frames over SPI at 30 FPS
Purpose	Offloads face rendering from Pi Zero 2W — Pi CPU free for Gemini + OpenCV

ROLE IN ADAM v32: Dedicated face rendering GPU. Receives emotion state strings via UART (e.g. 'happy', 'sad', 'thinking') from ESP32-CAM and renders matching animations on the ILI9341 TFT at 30 FPS. This completely decouples face animation from the AI processing loop on the Pi Zero 2W.

SOFTWARE NOTE: `adam_face.html` WebSocket renderer is deprecated for the physical unit. A new MicroPython SPI TFT firmware must be written for the Pico (port of emotion animation logic). WebSocket approach remains valid for the web demo / developer preview.

Price (INR): Rs. 299 - Rs. 399 — robu.in

C04 ILI9341 TFT Display 2.42" (240x320 SPI)

NEW v32

ADAM's Face Display — Replaces SSD1309 OLED

Driver IC	ILI9341 (SPI interface)
Resolution	240 x 320 pixels — full colour RGB565
Colour	Full 65K colour — enables colour emotion animations (not white-only)
Interface	SPI 4-wire: MOSI GP19, SCLK GP18, CS GP17, DC GP16, RST GP20
Supply	VCC + LED → Pico 3V3(OUT) Pin 36 — CRITICAL: 3.3V ONLY, never 5V
Controller	Driven exclusively by Raspberry Pi Pico over SPI

CRITICAL POWER: ILI9341 VCC and LED backlight pins MUST connect to Pico 3V3(OUT). Feeding 5V directly will permanently damage the display logic. The Pico's clean 3.3V rail is sufficient for the display's ~25-35 mA draw.

DISPLAY UPGRADE: Replaces SSD1309 OLED (128x64 monochrome). ILI9341 offers 240x320 full-colour, enabling richer emotion animations with colour coding. OLED bezel design must be updated for new display dimensions.

Price (INR): Rs. 299 - Rs. 399 — robu.in

C06 MG90S Metal Gear Micro Servo — Qty: 2

ORIGINAL

Servo Neck — Pan (Body, Left/Right) + Tilt (Head, Up/Down)

Gear type	Full metal gears — prevents stripping under continuous micro-movements
Torque	2.2 kg/cm @ 5V — sufficient for ADAM head weight
Speed	0.09 sec / 60 degrees — fast enough for reactive face tracking
Pan signal	PWM 50 Hz — Pi Zero 2W GPIO12 (was GPIO9 in v2.2)
Tilt signal	PWM 50 Hz — ESP32-CAM GPIO13 (moved off Pi entirely in v32)
Supply	4.8V - 6V — ISOLATED 5V RAIL with 470uF caps — NEVER from Pi or ESP32 pins

CRITICAL: Both servos powered from ISOLATED 5V rail sharing only GND with Pi. Place 470uF capacitors physically at each servo connector. Pan servo lives in the body; tilt servo lives in the head.

Price (INR): Rs. 179 - Rs. 229 each (x2 = Rs. 358 - Rs. 458) — robu.in

C07 TTP223 Touch Sensors x4 + PCF8574T I2C Expander

UPDATED

Touch Input — Silent Capacitive Interaction, I2C-Grouped

TTP223 IC	TTP223-BA6 capacitive IC — digital HIGH/LOW output (active HIGH)
Supply	2.0V - 5.5V, ~1.5uA standby — ESP32-CAM 3.3V
Penetration	Works through 3-5 mm PLA/acrylic — flush-mount compatible
PCF8574T	I2C expander — maps 4 touch sensors to 2 I2C pins (ESP32 GPIO14/15)
Mapping	P0=Touch1, P1=Touch2, P2=Touch3, P3=Touch4
Functions	Wake/Attention, Mute mic, Cycle emotion, Sleep toggle

v32 CHANGE: Touch sensors moved from Pi GPIO pins to PCF8574T expander on ESP32-CAM. This frees Pi GPIO pins and keeps all head I/O on the ESP32-CAM. PCF8574T and TTP223 sensors all powered from ESP32-CAM 3.3V.

Price (INR): TTP223 Rs. 45-65 each (x4) + PCF8574T Rs. 89 — robu.in

C08 INMP441 I2S Microphone — Qty: 2 (Stereo)

UPDATED

Audio Input — Stereo ADAM Ears for Gemini Live Voice Capture

Interface	I2S digital — BCK, WS/LRCK, SD (no analog noise path)
SNR	61 dB — clear voice at desk distance
Supply	1.8V - 3.3V, ~1.4 mA each — CRITICAL: Pi 3.3V ONLY (never 5V)
Sample Rate	16 kHz — matches Gemini Live input format, no resampling needed
Channel	Mic 1: L/R tied to GND (left). Mic 2: L/R tied to 3.3V (right)
Wiring	Both mics share BCK GPIO18, WS GPIO19. Mic1 SD=GPIO20, Mic2 SD=GPIO20 (shared SD bus)

v32 CHANGE: Upgraded from mono (x1) to stereo (x2). Both INMP441 mics share the same I2S clock lines (GPIO18/19) on the Pi. Left/Right channel selection via the L/R pin — tie to GND for left channel, 3.3V for right. CRITICAL: Both mics must use Pi 3.3V, never 5V.

Price (INR): Rs. 149 - Rs. 199 each (x2 = Rs. 298 - Rs. 398) — robu.in

C09 MAX98357A I2S DAC + Class-D Amplifier

SELECTED

Selected over PAM8403: single I2S chip, better SNR (77 dB), simpler wiring, same 3W output.

Interface	I2S: BCK GPIO18, LRCLK GPIO19, DIN GPIO21 — direct from Pi
Output	3.2W @ 4 Ohm, 5V mono — matches 3W speaker perfectly
SNR	77 dB — better than PAM8403 analog path
Supply	2.7V - 5.5V — from 5V Main Rail
ALSA	dtoverlay=hifiberry-dac in /boot/config.txt (or /boot/firmware/config.txt)

C11 — 3W 4-Ohm Speaker: 40mm bottom-firing, 200 Hz - 18 kHz. Placed at body bottom for omnidirectional desk projection. Perfect impedance match for MAX98357A.

Amp Price: Rs. 199 - Rs. 249 | Speaker: Rs. 79 - Rs. 99 — robu.in

C12 RCWL-0516 Microwave Radar Module

NEW v32

Presence Detection — Replaces VL53L0X ToF Sensor

Technology	RCWL-9196 Doppler microwave radar — detects motion in 360°
Range	Up to ~7m detection radius — ideal room-scale presence wake
Output	Digital HIGH when motion detected — GPIO16 on Pi Zero 2W
Supply	4V - 28V (5V from Main Rail) — has internal 3.3V regulator
Current	~2.8 mA typical — very low standby power
vs VL53L0X	No I2C polling needed — single digital pin, simpler code, longer range

ROLE IN ADAM v32: Wake ADAM when someone enters the room. Mounted inside the plastic body (microwaves pass through PLA/acrylic). Pi reads GPIO16: HIGH = someone present, trigger attention state. Replaces proximity-distance logic of VL53L0X; no more 80cm/20cm thresholds — just wake/sleep toggle.

SOFTWARE NOTE: Replace VL53L0X I2C polling loop with a GPIO interrupt on GPIO16. `RPi.GPIO.add_event_detect(16, GPIO.RISING, callback=presence_callback)`.

Price (INR): Rs. 79 - Rs. 119 — robu.in / electronicscomp.com

C13 / C14 / C15 Power Distribution

UPDATED

C13 Buck Conv.	LM2596 or MP1584 — 5V output, 3A minimum rated — from 2S LiPo 7.4V input
C14 Battery	2S LiPo 7.4V 2000mAh — fits inside 95mm body diameter
C15 Power Mgmt	AMS1117-3.3 LDO (for legacy if needed), 1000uF bulk cap on 5V bus, 470uF x2 at servo connectors, 1N5819 Schottky reverse-polarity, polyfuse 2.2A trip
RGB LED	5mm common cathode, GPIO17/27/22 — status: slow white=idle, blue=attention, cyan blink=API, green=memory, red=error

Buck Converter: Rs. 99-199 | LiPo 2000mAh: Rs. 499-699 — robu.in / amazon.in

C16 Micro-SD Card — 32GB Class 10 / A1

ORIGINAL

Capacity	32GB minimum — Vosk model ~1GB, OS ~4GB, ample headroom
Speed	Class 10 / A1 rated — A1 for low-latency random I/O
OS	Raspberry Pi OS Lite — headless, SSH enabled

Price (INR): Rs. 249 - Rs. 399 (32GB A1) — amazon.in (Samsung/SanDisk)

5. MASTER PINOUT CONNECTIONS (v32)

Complete pinout for all four MCUs. Physical pin numbers are given for Pi Zero 2W; GPIO numbers for ESP32-CAM and Pico. Colour coding: **BODY** = green, **HEAD** = blue.

A. Raspberry Pi Zero 2W (Body)

Pin / GPIO	Connects To
Pin 12 / GPIO18	I2S BCK — Mic1+Mic2 BCK AND Amp BCLK (shared clock)
Pin 35 / GPIO19	I2S LRC — Mic1+Mic2 WS AND Amp LRCLK (shared clock)
Pin 38 / GPIO20	I2S Data IN — Mic SD (both mics share, L/R pin selects channel)
Pin 40 / GPIO21	I2S Data OUT — Amp DIN
Pin 32 / GPIO12	Pan Servo PWM signal (Orange/Yellow wire) ← changed from GPIO9
Pin 36 / GPIO16	RCWL-0516 Radar OUT ← replaces VL53L0X I2C
Pin 8 / GPIO14 (TX)	FLOW 2 ONLY: to ESP32-CAM GPIO3 (U0RXD)
Pin 10 / GPIO15 (RX)	FLOW 2 ONLY: to ESP32-CAM GPIO1 (U0TXD)

B. ESP32-CAM (Head)

Pin / GPIO	Connects To
GPIO14	PCF8574T I2C Expander SDA
GPIO15	PCF8574T I2C Expander SCL
GPIO13	Tilt Servo PWM signal (Orange/Yellow wire)
GPIO4 (SW TX)	Pico GP1 (UART RX) — emotion command relay inside head
GPIO1 (U0TXD)	FLOW 2 ONLY: to Pi Zero GPIO15 (RX)
GPIO3 (U0RXD)	FLOW 2 ONLY: to Pi Zero GPIO14 (TX)

C. PCF8574T I2C Expander (Head)

Pin / GPIO	Connects To
SDA / SCL	ESP32-CAM GPIO14 / GPIO15
P0	TTP223 Touch Sensor 1 Signal (Wake/Attention)
P1	TTP223 Touch Sensor 2 Signal (Mute Mic)
P2	TTP223 Touch Sensor 3 Signal (Cycle Emotion)
P3	TTP223 Touch Sensor 4 Signal (Sleep Toggle)
VCC	ESP32-CAM 3.3V — keeps I2C logic levels at 3.3V

D. Raspberry Pi Pico (Head)

Pico is powered by the Head 5V Rail → VSYS (Pin 39). The TFT display is powered by Pico 3V3(OUT) Pin 36 at 3.3V. Never connect TFT VCC to 5V.

Pin / GPIO	Connects To
GP19 (Pin 25)	TFT MOSI
GP18 (Pin 24)	TFT SCLK / SCK
GP17 (Pin 22)	TFT CS
GP16 (Pin 21)	TFT DC
GP20 (Pin 26)	TFT RST
GP1 (Pin 2)	UART RX — receives emotion commands from ESP32-CAM GPIO4
VSYS (Pin 39)	Power IN — from Head 5V Rail (5V)
3V3(OUT) Pin 36	Power OUT — to ILI9341 TFT VCC + LED (3.3V)

6. ENCLOSURE & MECHANICAL HARDWARE

Enclosure dimensions unchanged from v2.2. Head bezel must be redesigned for ILI9341 TFT (240x320, PCB ~54x86mm) vs old SSD1309 OLED (PCB 72x43mm). Body and neck scoop geometry unchanged.

Parameter	Spec	Notes
Body Diameter	95 mm outer	2mm wall thickness
Body Height	90 mm	Foot flare R28, neck scoop R22
Head	104 x 75.5 mm oval	Bezel redesign needed for ILI9341 TFT PCB size
Material	PLA+ (recommended)	Better layer adhesion vs standard PLA
Layer Height	0.15 - 0.20 mm	0.15mm for display bezel, 0.20mm for body
Infill	25% Gyroid	Even strength in all axes
Supports	Tree supports	Servo pockets + neck scoop require supports
Colour	Matte Black PLA+	Matches DGEN brand aesthetic
Black Acrylic	3mm, A4 sheet	Side panel / decorative fascia
Brass Inserts	M3 x4mm OD	Heat-set with soldering iron at ~200°C
Countersunk Screws	M3 Countersunk x5	Flush panel: 2x front, 2x rear, 1x base

M01 — Black Acrylic Sheet 3mm (A4)

Rs. 160-200

Decorative side panel. Cut with laser cutter or scroll saw. Bonds to PLA+ with cyanoacrylate gel.

M02 — M3 x4mm OD Brass Heat-Set Inserts (x20)

Rs. 109-130

Install with soldering iron at ~200°C. Prevents thread stripping across repeated disassembly cycles.

M03 — M3 Countersunk Screws x5

Rs. 49-69

5x flush-mount positions: 2x front acrylic, 2x rear panel, 1x base plate. All others use pan-head M3.

M06 — Misc Consumables

Rs. 150

Solder (60/40 rosin core), 22AWG wire (red/black/colours), heatshrink, rosin flux pen, insulating tape. Rash Behari Avenue or Hatibagan Electronics Market, Kolkata.

C18 — Heat Sinks

Rs. 79

Self-adhesive aluminium. Apply to Pi Zero 2W SoC, ESP32 chip, and Pico RP2040 chip. Clean with IPA first. Leave 30 min cure time before full load.

7. GPIO PIN ASSIGNMENTS — Pi Zero 2W (Body)

Pin assignments for the Pi Zero 2W body MCU in v32. Tilt servo and touch sensors have moved to the ESP32-CAM; I2C bus is now fully free on the Pi (OLED and VL53L0X removed).

GPIO PIN	Function	Component	Protocol
GPIO18 (PCM_CLK)	I2S Bit Clock	INMP441 Mic x2 + MAX98357A (shared clock)	I2S
GPIO19 (PCM_FS)	I2S LR Clock	INMP441 Mic x2 + MAX98357A (shared clock)	I2S
GPIO20 (PCM_DIN)	I2S Data In	INMP441 Mic x2 (L/R pin selects channel)	I2S
GPIO21 (PCM_DOUT)	I2S Data Out	MAX98357A DAC Amp	I2S
GPIO12	PWM Pan Servo	MG90S Pan — Left/Right ← NEW (was GPIO9)	PWM 50Hz
GPIO16	Digital Input	RCWL-0516 Radar OUT ← NEW (replaces VL53L0X)	GPIO IRQ
GPIO14 (UART TX)	UART TX	Flow 2 only: ESP32-CAM GPIO3 (U0RXD)	UART
GPIO15 (UART RX)	UART RX	Flow 2 only: ESP32-CAM GPIO1 (U0TXD)	UART
GPIO17/27/22	Digital Output	RGB Status LED (R/G/B)	GPIO PWM
GPIO2/3 (I2C)	FREE in v32	Previously: OLED + VL53L0X — now available	—
GPIO9/10	FREE in v32	Previously: Pan/Tilt servos — now available	—
GPIO5/6/13/26	FREE in v32	Previously: TTP223 touch — moved to ESP32	—

ESP32-CAM (Head) Pin Summary

GPIO PIN	Function	Component	Protocol
GPIO14	I2C SDA	PCF8574T Touch Expander	I2C
GPIO15	I2C SCL	PCF8574T Touch Expander	I2C
GPIO13	PWM Tilt	MG90S Tilt Servo ← NEW in v32	PWM 50Hz
GPIO4 (SW TX)	UART TX	Pico GP1 — emotion relay	UART
GPIO1 (U0TXD)	UART TX	Flow 2: Pi GPIO15 (RX)	UART
GPIO3 (U0RXD)	UART RX	Flow 2: Pi GPIO14 (TX)	UART

I2S shared clock note: GPIO18 (BCK) and GPIO19 (LRCLK) are shared between both INMP441 mics and the MAX98357A. This is standard Pi I2S bus topology — both devices use the same clock. The L/R pin on each mic selects its channel (GND=left, 3.3V=right). Confirmed working with `dtoverlay=hifiberry-dac + sounddevice` library.

8. SHOPPING GUIDE — WHERE TO BUY IN INDIA

All sites sell individual units (no minimum order). Verified Indian sellers, May 2026. robu.in ships from Bengaluru — typically 3-5 days to Kolkata.

Component	Price (Rs.)	Site	Search / Notes
Raspberry Pi Zero 2W	1,099-1,299	robu.in	Search: 'Pi Zero 2W' — with headers version only
ESP32-CAM Module	349-499	robu.in	Search: 'ESP32-CAM' — AI Thinker variant (OV2640)
Raspberry Pi Pico	299-399	robu.in	Search: 'Raspberry Pi Pico' — standard (not W)
ILI9341 TFT 2.42"	299-399	robu.in	Search: 'ILI9341 2.4 SPI' — 240x320 colour
MG90S Servo (x2)	179-229 each	robu.in	Search: 'MG90S metal gear' — metal gear ONLY
TTP223 Touch (x4)	45-65 each	robu.in	Search: 'TTP223' — digital output module
PCF8574T I2C Expander	79-119	robu.in	Search: 'PCF8574T' or 'PCF8574 I2C expander'
INMP441 Mic (x2)	149-199 each	robu.in	Search: 'INMP441' — I2S breakout module
MAX98357A DAC/Amp	199-249	robu.in	Search: 'MAX98357A' — I2S, 3W mono
3W 4-Ohm Speaker	79-99	robu.in	Search: '40mm 3W speaker' — 4 Ohm impedance
RCWL-0516 Radar	79-119	robu.in	Search: 'RCWL-0516' — microwave radar module
5V Buck Converter 3A	99-199	robu.in	Search: 'LM2596 buck' or 'MP1584 5V' — 3A min
2S LiPo 7.4V 2000mAh	499-699	amazon.in	Search: '2S LiPo 7.4V 2000mAh' — with JST connector
32GB A1 microSD	249-399	amazon.in	Samsung/SanDisk 32GB A1 — A1 rating mandatory
PLA+ Filament Black	600-800/kg	amazon.in	eSUN PLA+ Matte Black — ~300g per ADAM print
Black Acrylic 3mm A4	160-200	amazon.in	Search: 'black acrylic sheet 3mm A4'
M3 Brass Inserts x20	109-130	amazon.in	Search: 'M3 brass heat set inserts'
M3 Countersunk x5	49-69	amazon.in	Search: 'M3 countersunk stainless screws'
Heat Sinks (set)	79	amazon.in	Search: 'raspberry pi zero heatsink kit'
Misc Consumables	150	Local	Solder, wire, flux, heatshrink — Rash Behari/Hatibagan

Also check: electronicscomp.com, evelta.com, and thingbits.net for alternative electronics sourcing. amazon.in for mechanical and storage items (Prime delivery). Local Kolkata markets (Rash Behari Avenue, Hatibagan Electronics Market) for consumables — cheaper, same-day.

9. ASSEMBLY NOTES & CRITICAL WARNINGS

CRITICAL: SERVO POWER ISOLATION

NEVER power MG90S servos from Pi GPIO 5V pin or ESP32-CAM 5V pins. Stall currents (220mA+ each) WILL cause brownout resets and SD card corruption. Use dedicated 5V isolated rail sharing only GND. Place 470uF capacitors physically adjacent to each servo connector.

CRITICAL: ILI9341 TFT POWER

ILI9341 VCC and LED backlight MUST connect to Pico 3V3(OUT) — Pin 36. Never feed 5V. 5V will permanently damage the display logic. The Pico's 3.3V output is sufficient.

CRITICAL: INMP441 MIC POWER

Both INMP441 mics must use Pi Zero 2W 3.3V ONLY. Never use 5V — it will damage the Pi's I2S data pins. Both mics share GPIO18/19 clock lines. L/R pin: GND=left channel, 3.3V=right channel.

CRITICAL: COMMON GROUND (GOLDEN RULE)

All devices — Pi, ESP32-CAM, Pico, Amplifier, Servos, Buck Converter — MUST share a common GND. Without shared ground, I2S audio will produce noise and UART serial communication will fail.

CRITICAL: SD CARD WRITE SAFETY

Never pull power while ADAM is writing to adam_memory.json or adam_conversations.json. Implement graceful shutdown on SIGINT/SIGTERM that flushes all open file handles before power off.

CRITICAL: PICO FACE FIRMWARE

adam_face.html WebSocket renderer is deprecated for the physical unit. New MicroPython SPI TFT firmware must be written for the Pico. WebSocket approach remains valid for web demo only.

CRITICAL: M3 BRASS INSERTS

Require ~200°C soldering iron. Press slowly and straight — crooked inserts cause screws to bind. Let cool 45 seconds before threading. 5 countersunk positions: front panel x2, rear panel x2, base x1.

CRITICAL: ASYNCIO THREAD SAFETY

All servo pan(), tilt(), and emotion_move() calls MUST use asyncio.to_thread(). Direct servo calls inside async functions block the event loop and cause audio stuttering.

CRITICAL: PAN GPIO CHANGE

Pan servo moved from GPIO9 (v2.2) to GPIO12 (v32). Update all firmware and adam_neck_serial.py accordingly. Tilt servo now on ESP32-CAM GPIO13 — no longer driven by the Pi at all.

CRITICAL: I2S CLOCK SHARING

GPIO18 (BCK) and GPIO19 (LRCLK) are shared between both INMP441 mics and MAX98357A. Test audio output with aplay BEFORE wiring mics, to confirm DAC is working first.

Recommended Assembly Sequence

Step	Action	Verify Before Proceeding
1	Flash Pi Zero 2W SD card with Raspberry Pi OS Lite. Enable SSH + Wi-Fi.	SSH login works, pip install completes
2	Apply heat sinks: Pi Zero 2W SoC, ESP32 chip, Pico RP2040 chip.	Heat sinks seated flat, no capacitor contact
3	Flash ESP32-CAM firmware (Arduino IDE). Verify camera stream in browser.	Camera stream visible at ESP32-CAM IP
4	Flash Pico MicroPython firmware. Test ILI9341 TFT renders test pattern.	Display shows colour test pattern at 30 FPS
5	Wire MAX98357A I2S DAC (GPIO18/19/21). Configure dtoverlay. Test with aplay.	Speaker produces clean tone from aplay test
6	Wire both INMP441 mics (GPIO18/19/20, L/R channels). Test with arecord at 16kHz.	Clean stereo audio capture, minimal noise floor
7	Wire RCWL-0516 radar to GPIO16. Test digital output with GPIO.input(16).	GPIO16 goes HIGH when hand moves in front
8	Wire pan servo to ISOLATED 5V rail + GPIO12. Wire tilt servo to ESP32 GPIO13. Test sweeps.	No Pi reboot on stall, no ESP32 crash on stall
9	Wire PCF8574T expander to ESP32 GPIO14/15. Wire TTP223 sensors to P0-P3.	i2cdetect on ESP32 shows 0x20 (PCF8574T default)

10	Wire Pico GP1 (RX) to ESP32-CAM GPIO4 (SW TX). Send test emotion command.	<i>TFT shows correct emotion animation on command</i>
11	Heat-set M3 brass inserts into all PLA+ screw holes (200°C iron, 45s cool).	<i>All inserts flush, no wobble</i>
12	Mount all components. Route 2-wire (Flow 1) or 4-wire (Flow 2) through neck.	<i>All connectors accessible, no cable pinching</i>
13	Attach acrylic panel. Run full ADAM v32 Python suite.	<i>All subsystems init, Gemini responds, face animates</i>

